

DATA SCIENTIST

INTERVIEW CASE

Forecasting is an essential part of managing the supply chain and to ensure our products are always in stock. In BESTSELLER we have a range of tools to help planners maintain stock levels. In one of these tools, we need to forecast at a weekly granularity four weeks ahead what the demand is at SKU-level.

A SKU in BESTSELLER is a combination of a style, a colour and a size, e.g. [Round Neck Knitted Pullover, Black, L](#) or [Knitted Pencil Skirt, Natural Melange, XS](#).

1. We want to build an ML model to solve this forecasting problem.

What kind of model and predictor variables would you consider for this forecasting problem?

Given a specific style and colour, the above model will give forecasts for each size, and we can calculate the proportion each size has of a total. A priori, there is no mechanism to ensure these proportions are sensible, i.e., you get something like Figure 1 and not like Figure 2.

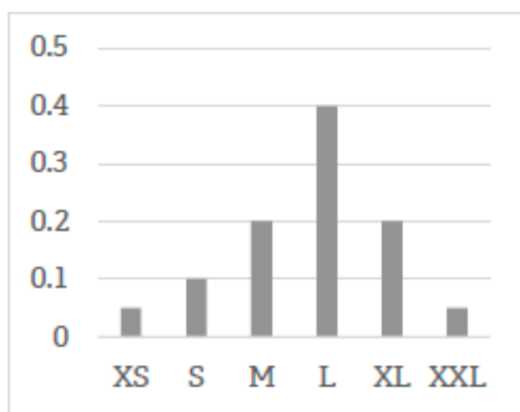


Figure 1: Sensibly looking size split of forecasts for a given style and colour.

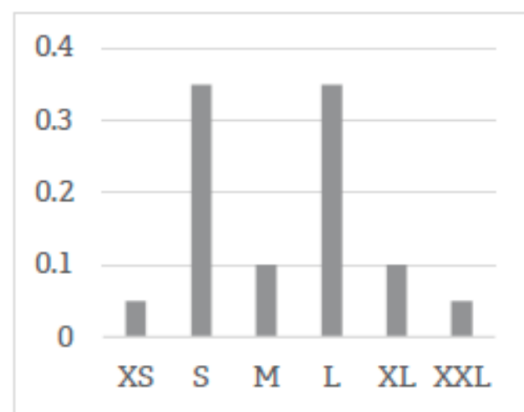


Figure 2: Incorrect size split of forecasts for a given style and colour.

2. What can be done to ensure our SKU-forecasts does not in combination give a wrong size split?

Stockouts represented in the historical sales data will be problematic for the model if not accounted for.

3. What options or strategies would you pursue to account for stockouts, so that we use SKU demand rather than SKU sales as target for our forecasting model?

The purpose of the exercise is to have a foundation for discussion. Please only use a couple of hours on this exercise – we are not looking for a fully detailed solution.

If there are any questions, please do reach out.